

Evaluation of Unsignalized Intersection at Bobergsgatan – Norra Fiskartorpsvägen



Course: AH2171 Traffic Engineering and Management

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1 Introduction

1.1 Background

The unsignalized T-intersection at Bobergsgatan and Norra Fiskartorpsvägen serves as a critical link between Östermalm, northern Stockholm, and Stockholm University. Figure 1.1 illustrates the location of the intersection, while Figure 1.2 presents the geometry of its current layout.

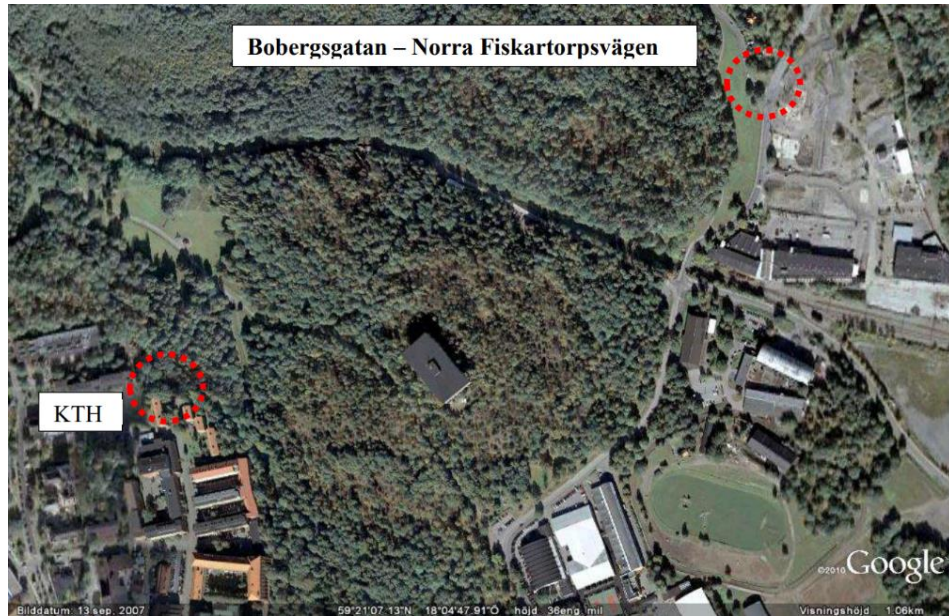


Figure 1.1 Intersection Location

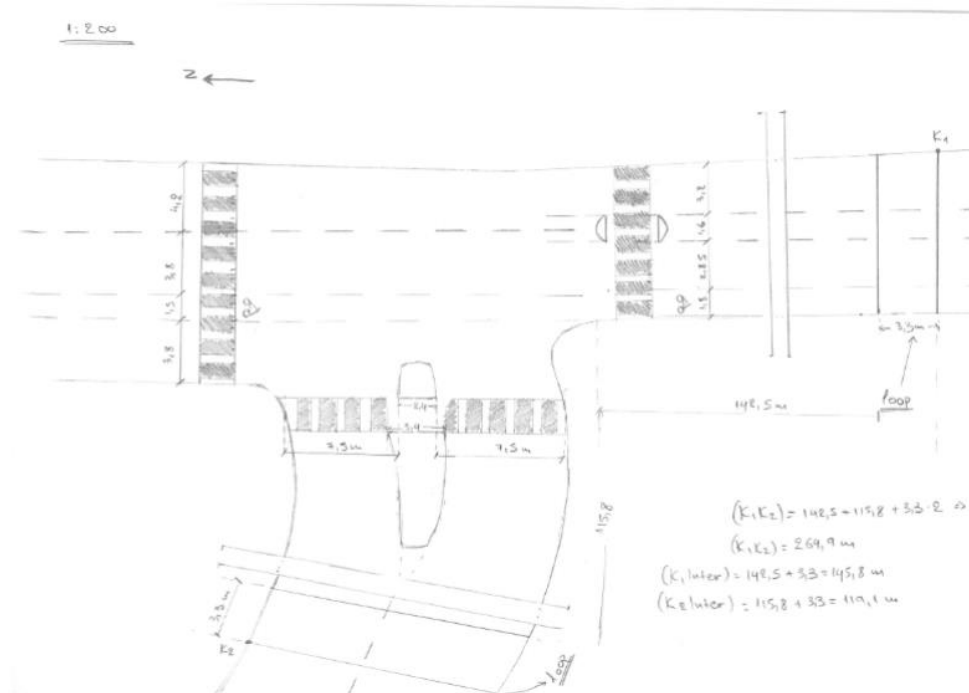


Figure 1.2 Geometry of the Intersection

With the growing demands of urban traffic, this intersection frequently experiences saturation during peak hours, raising concerns about its efficiency, safety, and overall level of service. In response to these challenges, it is essential to evaluate the intersection's performance and propose optimization strategies grounded in empirical data. Such efforts aim to enhance the reliability of the urban road network and promote sustainable transportation development.

1.2 Goals

This report aims to evaluate the performance of the unsignalized intersection at Bobergsgatan–Norra Fiskartorpsvägen through empirical research and quantitative analysis. The core objectives of the study are as follows:

1. By leveraging multi-source traffic data collection, the operational status of intersections can be accurately characterized.
2. Various analytical methods are employed to assess their current level of service.
3. Based on the evaluation results, feasible improvement plans are designed and compared to provide a solid foundation for traffic management decision-making.

1.3 Methods

To achieve the goals mentioned above, this report used multiple data collection methods, comprehensively used handheld pushbutton devices, pneumatic tubes, video cameras, and manual queue measurements gaining key traffic parameters such as traffic volumes, vehicle composition, turning movements, speeds, travel times, queue lengths, and delays. When analysing the data, direct measurement, the Highway Capacity Manual (HCM) 2010, and the CAPCAL method were used, conducting a multi-dimensional measurement and comparison of intersection delays and levels of service.

2 Method

2.1 Data Collection

Multiple methods were used to collect the data to evaluate the operational performance of the unsignalized intersection at Bobergsgatan and Norra Fiskartorpsvägen. The objective was to capture a complete dataset encompassing traffic volumes, vehicle composition, turning movements, speeds, travel times, and queue dynamics. To ensure data robustness and enable cross-validation, a combination of manual and automated data collection methods was deployed simultaneously at strategic locations within the study area.

2.1.1 Handheld Pushbutton Devices (Locations: PB1, PB2)

This method was employed to collect disaggregated data on traffic volumes and vehicle types for specific turning movements at intersection approaches, using handheld devices with four buttons connected to a portable data logger, primarily utilizing Buttons 1 and 2, where observers at predefined locations (PB1 and PB2) recorded vehicle passages by pressing the corresponding buttons based on type and manoeuvre; each event was timestamped and

tagged with a channel number indicating the movement and vehicle type, resulting in a raw data stream of sequential records over event type, channel number, and timestamp.

2.1.2 Manual Queue Length Measurement (Locations: Q1, Q2)

To quantify the spatiotemporal extent of congestion, vehicle queue lengths were measured at critical points using a structured observation method. At each designated location (Q1 and Q2), a team of two observers was deployed: one observer signalled 30-second intervals, while the other simultaneously counted and recorded the number of vehicles in the queue during each interval. The resulting data was compiled into a time-series table, detailing each observation period alongside the corresponding number of queued vehicles (N_i), enabling precise temporal tracking of congestion levels.

2.1.3 Geometry Survey

To obtain precise physical dimensions of the intersection for analysis and modelling, measurements were conducted using a measuring wheel and manual sketching tools. The distance between key points, most notably from K1 to K2, was recorded, along with comprehensive geometric data such as lane widths, lengths of auxiliary lanes, and the dimensions of islands and other physical features. This information provided a detailed spatial foundation for subsequent modelling and evaluation.

2.1.4 Pneumatic Tube Detectors (Locations: K1, K2)

To automatically and continuously record traffic volume, vehicle classification, and spot speed at two key points flanking the intersection, pneumatic tubes were installed across the travel lanes and connected to a TMS data logger. As vehicles passed over the tubes, air pulses were generated by each axle and recorded. The time intervals between axle crossings were used to calculate vehicle speed, determine axle count, and classify vehicle types. The proprietary API software processed these signals into structured data for each vehicle, including a sequential vehicle number, timestamp of the first axle hit, travel channel and direction, vehicle type (e.g., car, van, lorry), speed (in km/h), and number of axles.

2.1.5 Video Recording (Locations: V1, V2)

To capture vehicle trajectories and estimate travel time between points K1 and K2, remotely controlled video cameras were installed at both locations alongside the pneumatic tubes. These cameras recorded the front of each vehicle as it passed the detection points. The footage was later digitized and manually processed to match license plates across the two sites. For each vehicle observed within the designated time window, the following data was recorded in a spreadsheet: license plate number, vehicle type and color, and the exact passage time marked by the moment the front bumper crossed the reference line.

2.1.6 Supplementary Data

To document contextual factors that could influence quantitative results, observers continuously recorded relevant observations throughout the data collection and video review process. This included noting any unusual events, prevailing weather conditions, and general site characteristics that might account for anomalies or patterns in the data. These qualitative

insights provided essential context for interpreting the measured outcomes and ensuring the reliability of subsequent analyses.

2.2 Data Processing

2.2.1 The Handheld Pushbutton Devices

Raw data collected from handheld pushbutton devices was transferred to an Excel file. Data recorded between 14:45 and 15:15 was retained, and the number of different vehicle types along with their turning directions was counted at 3 approaches A, D, and F in 5-minute intervals.

2.2.2 Pneumatic Tube Detectors (Locations: K1, K2)

To process the gathered information in ASCII-format an API software is used to record the parameters: tube number, passage times, travel direction, number of axles and distances, and vehicle types and speeds. The software generates two output files containing aggregated data according to a specific time period, 5 and 15 min.

2.2.3 Video Recording (Locations: V1, V2)

The digitalised video recordings at locations V1 and V2 are manually processed to collect information about the plate numbers of the vehicles at locations V1(K1) and V2(K2). By matching the plate numbers of vehicles that have crossed both locations, the travel between the two locations can be estimated.

2.2.4 Other Data

The queue data collected at Q1, Q2 were already processed into an Excel file. And the geometrical data of the intersection is shown in the Figure 1.2.

3 Results

3.1 Processing the Data from the Handheld Pushbutton Devices

Tables 3.1 to 3.3 present summaries of traffic flows at approaches A, D, and F.

Table 3.1 Summary of Traffic Flow at Approach A

Flow turning movements-A		
Time Intervals	Straight	Right turn
14:45-14:50	3	42
14:50-14:55	5	46
14:55-15:00	3	38
15:00-15:05	8	56
15:05-15:10	10	50
15:10-15:15	8	42

Table 3.2 Summary of Traffic Flow at Approach F

Flow turning movements-F		
Time Intervals	Left turn	Straight
14:45-14:50	29	11
14:50-14:55	20	6
14:55-15:00	12	5
15:00-15:05	22	7
15:05-15:10	15	4
15:10-15:15	23	6

Table 3.3 Summary of Traffic Flow at Approach D

Flow turning movements-D		
Time Intervals	Left turn	Right turn
14:45-14:50	26	7
14:50-14:55	15	16
14:55-15:00	20	8
15:00-15:05	23	9
15:05-15:10	31	15
15:10-15:15	32	8

3.2 Pneumatic Tube Detectors (Locations: K1, K2)

The processed data are grouped into 5-minute intervals from 13:45 – 14:45, making it a total of 12-time intervals. Within the 5 minutes interval the number of vehicles and average speed is calculated for each location. The processing data for K1 are shown in Table 3.4

Table 3.4: Processing data for K1

Time interval	Time	Travel Direction	Number of vehicles	Average speed at K1
1	13:45 - 13:50	1-0	19	36.8
2	13:50 - 13:55	1-0	14	30.6
3	13:55 - 14:00	1-0	13	35.6
4	14:00 - 14:05	1-0	21	34.7
5	14:05 - 14:10	1-0	10	32.5
6	14:10 - 14:15	1-0	17	32.6
7	14:15 - 14:20	1-0	14	36.6
8	14:20 - 14:25	1-0	14	33.7
9	14:25 - 14:30	1-0	20	33.9
10	14:30 - 14:35	1-0	22	33.8
11	14:35 - 14:40	1-0	17	34.2
12	14:40 - 14:45	1-0	19	32.3

The processing data for K2 is shown in Table 3.5 below.

Table 3.5: Processing data for K2

Time interval	Time	Travel Direction	Number of vehicles	Average speed at K2
1	13:45 - 13:50	0-1	34	34.5
2	13:50 - 13:55	0-1	37	32.1
3	13:55 - 14:00	0-1	31	35.2
4	14:00 - 14:05	0-1	25	29.9
5	14:05 - 14:10	0-1	32	35.5
6	14:10 - 14:15	0-1	38	30.9
7	14:15 - 14:20	0-1	31	34.9
8	14:20 - 14:25	0-1	32	33.5
9	14:25 - 14:30	0-1	40	31.1
10	14:30 - 14:35	0-1	34	33.9
11	14:35 - 14:40	0-1	40	29.4
12	14:40 - 14:45	0-1	52	29.6

Figures 3.1 and 3.2 illustrates the average speed and vehicle flow at the locations K1 and K2 per 5 minutes interval.

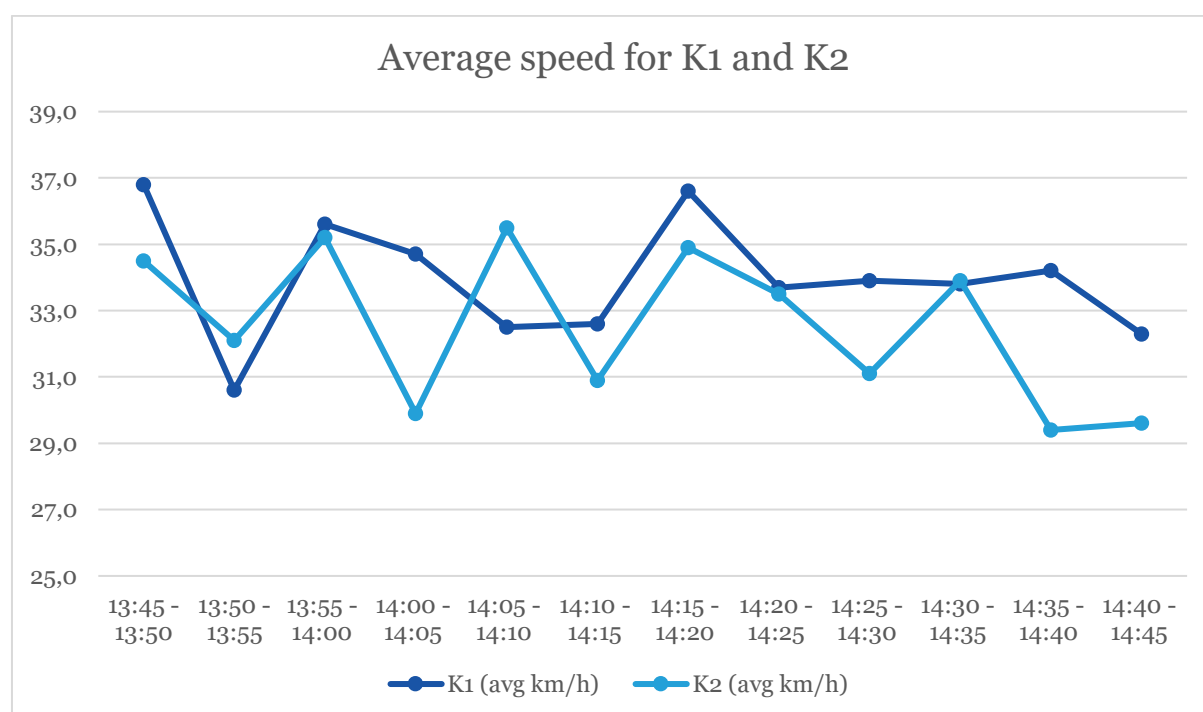


Figure 3.1: Average speed for K1 and K2

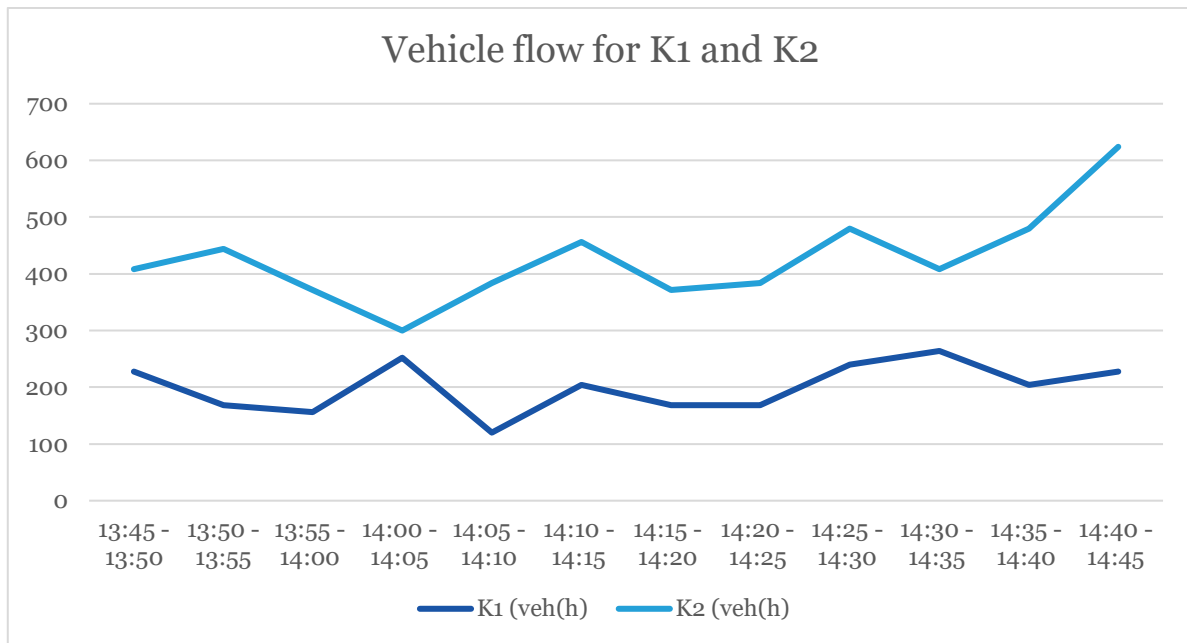


Figure 3.2 Vehicle flow for K1 and K2

Given the data the peak hour factor was calculated and is shown in Table 3.6.

Table 3.6: Peak hour factor per 15 minutes interval

Peak hour factor	
PHF (K1)	0.89
PHF (K2)	0.82

3.3 Video Recording (Locations: V1, V2)

Table 3.7 below shows the result from processed data from video cameras at locations V1 and V2 between 13:53:40 and 14:02:10. A total of 7 vehicles passed both K1 and K2.

Table 3.7 Matched licence plate number at K1 and K2 (13:53:40-14:02:10)

Vehicle plate license number	Vehicle type	Passage time at K1	Passage time at K2	Travel time K1-K2 (sec)
LHU703	car	13:54:43	13:55:18	35
BDT066	car	13:55:41	13:55:22	41
DRU146	truck	13:55:45	13:56:23	38
XXX3X7	car	13:55:46	13:56:24	38
LXH050	car	14:00:01	14:00:28	27
MNZ854	car	14:00:28	14:01:07	39
BUO245	car	14:00:56	14:01:27	31

4 Analysis

4.1 Exploratory Analysis

4.1.1 Location K1 and K2

Table 4.1 displays the combined percentage of different vehicle types for K1 and K2 per 5 minutes interval.

Table 4.1: Percentage of different vehicle types

Time	13:50	13:55	14:00	14:05	14:10	14:15	14:20	14:25	14:30	14:35	14:40	14:45
Private %	87.9	86.0	87.5	71.4	89.1	85.7	97.8	81.1	82.9	88.1	88.9	87.3
Heavy %	12.1	14.0	12.5	28.6	10.9	14.3	2.2	18.9	17.1	11.9	11.1	12.7

4.1.2 Location PB1 and PB2

The percentage of different vehicle types at the corresponding locations for PB1 and PB2 was calculated and that can be seen in Table 4.2 below.

Table 4.2: Vehicle compositions at the corresponding locations

Time Interval	From A to C		From A to B		From F to C		From F to E		From D to E		From D to B	
	Private vehicles	Heavy vehicles	Private vehicles	Heavy vehicles	Private vehicles	Heavy vehicles	Private vehicles	Heavy vehicles	Private vehicles	Heavy vehicles	Private vehicles	Heavy vehicles
14:45-14:50	100%	0%	100%	0%	100%	0%	90.9%	9.1%	92.3%	7.7%	85.7%	14.3%
14:50-14:55	100%	0%	80%	20%	100%	0%	66.7%	33.3%	87.5%	12.5%	66.7%	33.3%
14:55-15:00	100%	0%	100%	0%	100%	0%	100%	0%	85%	15%	100%	0%
15:00-15:05	100%	0%	75%	25%	100%	0%	100%	0%	100%	0%	66.7%	33.3%
15:05-15:10	96%	4%	90%	10%	100%	0%	25%	75%	93.5%	6.4%	86.7%	13.3%
15:10-15:15	100%	0%	87.5%	12.5%	100%	0%	66.6%	33.3%	90.6%	9.4%	100%	0%

Turning percentage of each section was calculated and can be seen in Table 4.3.

Table 4.3: Turning percentages for the period of analysis at the various locations

Time Interval	From A to C(Right Turning)	From A to B(Straight)	From F to C(Left turning)	From F to E(Straight)	From D to E(Left turning)	From D to B(Right turning)
14:45-14:50	93.33%	6.67%	72.50%	27.50%	78.79%	21.21%
14:50-14:55	90.20%	9.80%	76.92%	23.08%	51.61%	48.39%
14:55-15:00	92.68%	7.32%	70.59%	29.41%	71.43%	28.57%
15:00-15:05	87.50%	12.50%	75.86%	24.14%	71.90%	28.10%
15:05-15:10	83.33%	16.67%	78.95%	21.05%	67.39%	32.61%
15:10-15:15	84.00%	16.00%	79.31%	20.69%	80%	20%

The PHF for the total flow on the intersection was calculated and can be seen on Table 4.4 below. The volume is put into 15-minute sections and from that the PHF was found.

Table 4.4: PHF for the total flow of manual measurements

Time Interval	PB1 - Volume	PB2 - Volume	Combined
14:45-14:50	198	114	312
15:00-15:15	234	135	369
PHF	0.92		

4.2 Analysis of Studied Segment (K1 – K2)

4.2.1 Long-Base measured delay (K1 to K2)

Based on the geometric data of the intersection, the distance between points K1 and K2 is 264.9 meters. With an average posted speed limit of 40 km/h, the theoretical travel time without interference is calculated to be 24.8 seconds. The actual average travel time between K1 and K2, derived from video observations, is 38.79 seconds. Consequently, the delay caused by intersection interference is determined to be 13.79 seconds, as shown in Table 4.5.

Table 4.5: Delay on the intersection from interference

Average travel time - K1 to K2 (sec)
38.79
No interference travel time (sec)
24.80
Delay from interference (sec)
13.99

The average delay was calculated by comparing the observed travel time with a theoretical travel time under no-interference conditions, derived from either the speed limits along the segment or the measured average speed at K1 and K2. To calculate the speed limit per road to calculate the corresponding travel time (sec). The following equation is followed:

$$TT_{SL} = \frac{142.5+3.3}{\left(\frac{50}{3.6}\right)} + \frac{115.8+3.3}{\left(\frac{30}{3.6}\right)} \approx 24.79 \text{ s} \quad (1)$$

Using the average speed to calculate the to travel time can be calculated using this equation:

$$TT_{AvgV} = \frac{142.5+3.3}{\left(\frac{32.7}{3.6}\right)} + \frac{115.8+3.3}{\left(\frac{25.0}{3.6}\right)} \approx 32.18 \text{ s} \quad (2)$$

The results from equation one and two can now be utilized to calculate the control delay for both methods by using the average travel time (38.79 sec) and subtracting with the calculated values.

$$Delay_{SL} = 38.79 - 24.8 \approx 14.00 \text{ s} \quad (3)$$

$$Delay_{AvgV} = 38.79 - 32.3 \approx 5.61 \text{ s} \quad (4)$$

4.2.2 Direct Interference from queues (Q1 and Q2)

Figure 4.1 illustrates the variation in queue lengths over time at locations Q1 and Q2, recorded at 5-minute intervals throughout the observation period.

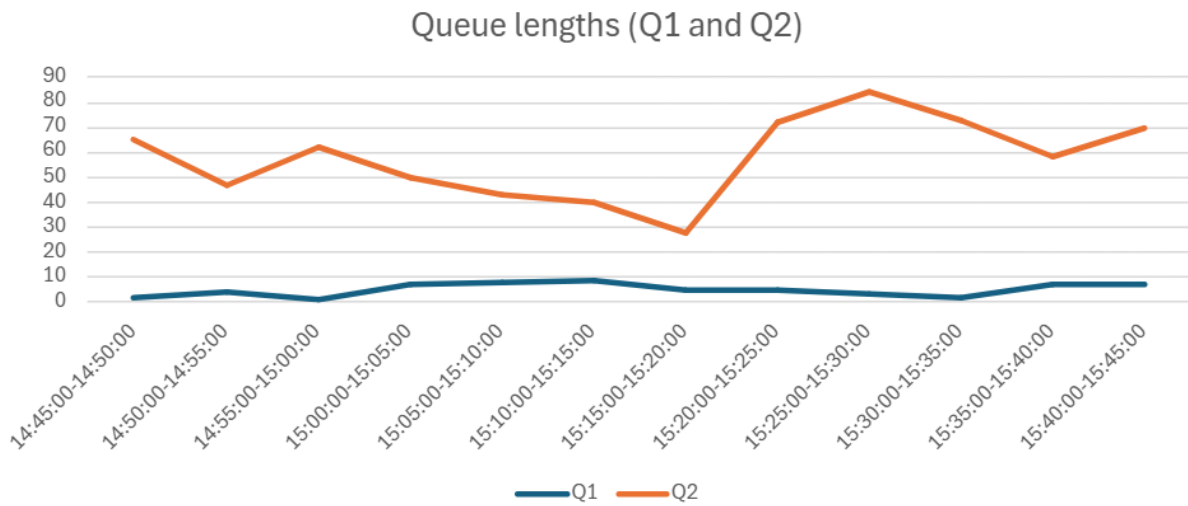


Figure 4.1 Queue lengths as a function of time at locations Q1 and Q2

The average delay per vehicle due to queuing at location Q1 is calculated to be 5.63 seconds per vehicle, based on Equation $d = \frac{1}{Q} \sum_{i=1}^{N(q)} n_i \cdot 30$. Since the pushbutton device data on departed vehicles covers only a 30-minute interval, while the study period spans one hour, the recorded number of departed vehicles, denoted as Q, is doubled to reflect the full duration. The resulting average delay calculations are presented in Table 4.5.

Table 4.5: Calculations of average delay per vehicle at Q1

Q	N(q)	SUM ni*30	Average delay per vehicle (sec/veh)
320	121	1800	5.63

4.3 Arrival with CAPCAL

Using the software CAPCAL, shown in Figure 4.2, the average delay per vehicle for the intersection was calculated by giving the input for geometry data, vehicle flow, private and heavy vehicle types.

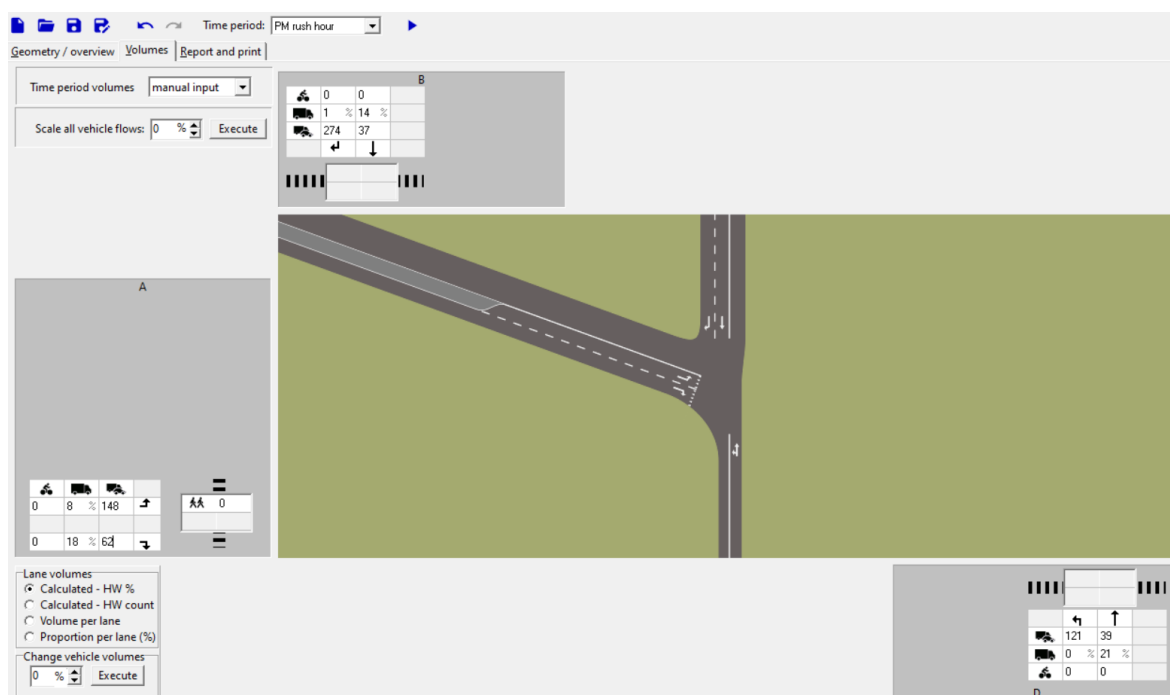


Figure 4.2 Intersection modelled in CAPCAL

Table 4.6 displays the result from Figure 4.2 that has been exported into an Excel file. In the table you see the delay and how the proportion of delayed vehicles in percentage. The abbreviations RT, LT, and THR for the column “movement” in the table stands for, right, left, straight.

Table 4.6 Average delay per vehicle on original intersection calculated by Capcal

Delay and number of stops per movement								
Approach	Movement	Delay (s/v)			Proportion of delayed vehicles %			Prop of veh that stops
		Conflict	Geom	Total	Conflict	Geom.	Total	
D	RT	3	3	5	7	93	100	3
	LT	5	4	8	38	62	100	24
	All	4	4	7	29	71	100	18
A	RT	0	2	2	0	100	100	0
	THR	0	0	0	0	0	0	0
	All	0	2	2	0	88	88	0
F	THR	0	1	1	10	9	18	1
	LT	3	4	4	27	73	100	9
	All	2	3	4	23	57	80	7

4.4 Arrival with HCM 2010

Table 4.7 below shows the results of all the important factors that were used to calculate the total delay. Only those turning lanes are shown since the other ones do not get any delay since they have the right of way on the intersection. Calculation in details can be found in the Appendix A.

Table 4.7 Calculated values from HCM 2010

	Left turning for major street (F-C)	Right turning for minor street (D-B)	Left turning for minor street (D-E)
Conflicting flow rate (veh/hr)	696	400	976
Critical headway (sec)	4.10	6.37	6.48
Follow-up headway (sec)	2.20	3.46	3.57
Potential capacity (veh/hr)	895	617	263
Movement capacity (veh/hr)	895	617	198
Control delay (sec)	10.50	12.36	393.40
LOS	B	B	F

4.5 Comparison of Different Methods

Four different methods were applied to estimate the average delay at the intersection: Long-Base measured delay, Direct Interference from queues, Capcal, and HCM 2010.

The Long-Base measured delay compares actual travel times with theoretical travel times under free-flow conditions. It was only calculated for vehicles traveling from K1 to K2, yielding an average delay of 13.99 seconds per vehicle.

The direct Interference from queues method uses queue counts at locations Q1 and Q2, recorded every 30 seconds, along with the number of vehicles departing within one hour. This produced an average delay of 5.63 seconds per vehicle.

The Capcal and HCM 2010 methods estimate delay for each approach and can also provide total intersection delay. For example, at approach F (left turn from Bobergsgatan to Norra Fiskartorpsvägen), Capcal calculated a delay of 4 seconds per vehicle, while HCM 2010 produced a value of 7.72 seconds per vehicle.

Because three of the methods only evaluate a single movement or approach, their results cannot be directly compared with the intersection-wide delays produced by Capcal. Table 4.8 summarizes the delay values for approach F across all four methods.

Table 4.8 Comparison of delay at approach F

	Long-Based measured delay	Direct interference from queues	Arrival with Capcal	Arrival with HCM 2010
Delay (s/veh)	13.99	5.63	4	7.72

The Long-Base measured delay is the only method that accounts for geometrical delay, yielding the highest value of 13.99 seconds per vehicle. While this makes it the most accurate representation of actual driver experience, its applicability is limited in scope. In comparison, the HCM 2010 method provides an estimated delay of 7.73 seconds per vehicle, which aligns most closely with the Long-Base results and is considered the most reliable approach for analyzing delay across the entire intersection.

5 Discussion

This part of the report, provides alternative design layouts to reduce the vehicle delay per seconds and then evaluates them to provide a final recommendation.

5.1 Alternative Designs

Figure 5.1 illustrates modifications were made to the original intersection using the lecture notes as guidance for developing an alternative design. The goal was to minimize the total delay and improve the efficiency of the intersection compared to the original layout. The key changes introduced in the new alternative design are a dedicated left turn for lane D in CAPCAL with the width of 3.2 meters resulting in a total width of 6.4 meters for lane D. The improvement with this design is that it reduces the waiting time for vehicles traveling straight in lane D. Additionally, all lanes now have their own dedicated turning lane.

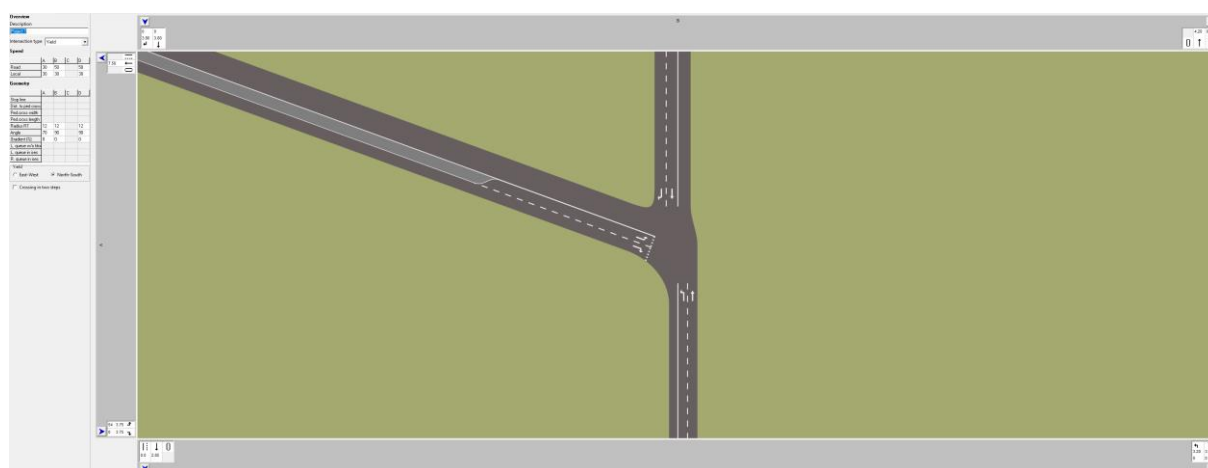


Figure 5.1: Two lane yield design layout

Table 5.1 the total delay for the intersection is 0.7 hours, approximately 42 minutes. This value is the sum of all individual vehicle delays across all intersection approaches. Contributing most

for the delays comes from approach A, left turn, followed by approach B, right turn. The degree and saturation remain low, and the queues stay below 0.3 vehicles; nearly all vehicles pass with minimum yielding time. Resulting in an average delay of 3.8 seconds per vehicle, indicating that the intersections operate with high efficiency and no signs of congestion.

Table 5.1: CAPCAL calculation results for two lane yield layout

Capacity and delay per lane						Queue length (num veh)	
Approach	Lane	Movement	Volume (vph)	Capacity (vph)	Degree of saturation	Mean	90th percentil
A	1	R	62	1205	0.05	0.1	0.1
	2	L	148	774	0.19	0.2	0.3
B	1	R	274	1992	0.14	0.0	0.0
	2	T	37	1765	0.02	0.0	0.0
D	1	T	39	1625	0.02	0.0	0.0
	2	L	121	1065	0.11	0.1	0.1

Delay and number of stops per lane								
		Delay (s/v)			Proportion of veh delayed			
Approach	Lane	Conflict	Geom.	Total	Conflict	Geom.	Total	Prop of veh that stops
A	1	3	3	5	7	93	100	3
	2	5	4	8	38	62	100	24
B	1	0	2	2	0	100	100	0
	2	0	0	0	0	0	0	0
D	1	0	0	0	0	0	0	0
	2	3	4	4	28	72	100	9
All vehicles		2	3	4	14	75	89	7
Delay and number of stops per movement								
		Delay (s/v)			Proportion of delayed vehicles %			
Approach	Movement	Conflict	Geom.	Total	Conflict	Geom.	Total	Prop of veh that stops
A	RT	3	3	5	7	93	100	3
	LT	5	4	8	38	62	100	24
	All	4	4	7	29	71	100	18
B	RT	0	2	2	0	100	100	0
	THR	0	0	0	0	0	0	0
	All	0	2	2	0	88	88	0
D	THR	0	0	0	0	0	0	0
	LT	3	4	4	28	72	100	9
	All	2	3	3	21	54	76	7
Total delay (hours)		0.7						

5.2 Roundabout

To replace the existing intersection, a roundabout design shown in Figure 5.2, is developed for further comparison with the other alternatives. The roundabout features two lanes of every approach except for lane D that has a single lane for both driving straight and turning left.

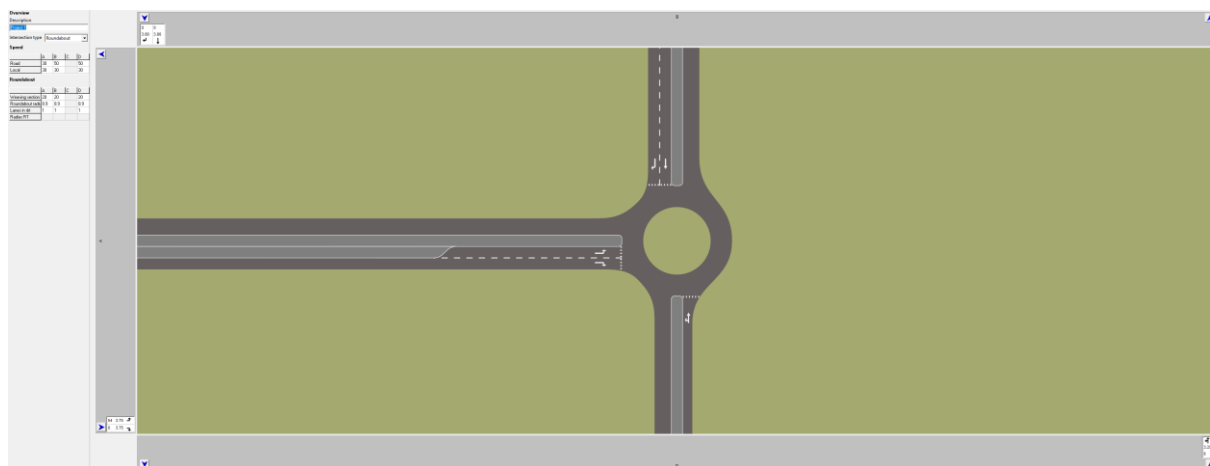


Figure 5.2: Roundabout layout

The results displayed in Table 5.2 shows that the roundabout operate with minimal delays. The roundabout solution minimises the conflict delay, however, creates geometric delay with 5-6 seconds per vehicle for right and left turning vehicles on approach A, and approaches B and D have similar delays. Due to short yielding times the resulting stop proportion remains close to zero even as there is a proportion of vehicles are classified as delay.

Table 5.2: CAPCAL results for roundabout layout

Capacity and delay per lane							Queue length (num veh)	
Approach	Lane	Movement	Volume (vph)	Capacity (vph)	Degree of saturation	Mean	90th percentile	
A	1	R	62	1383	0.04	0.0	0.0	
	2	L	148	1441	0.10	0.0	0.0	
B	1	R	274	1389	0.20	0.0	0.0	
	2	T	37	1291	0.03	0.0	0.0	
D	1	TL	160	1308	0.12	0.0	0.0	
Delay and number of stops per lane								
		Delay (s/v)			Proportion of veh delayed			
Approach	Lane	Conflict	Geom.	Total	Conflict	Geom.	Total	Prop of veh that stops
A	1	0	0	0	13	87	100	0
	2	0	6	6	13	87	100	0
B	1	1	0	0	25	75	100	0
	2	1	5	5	22	78	100	0
D	1	1	6	6	23	77	100	0
All vehicles		1	3	3	21	79	100	0

Delay and number of stops per movement								
Approach	Movement	Delay (s/v)		Total	Proportion of delayed vehicles %			Prop of veh that stops
		Conflict	Geom.		Conflict	Geom.	Total	
A	RT	0	0	0	13	87	100	0
	LT	0	6	6	13	87	100	0
	All	0	5	5	13	87	100	0
B	RT	1	0	0	25	75	100	0
	THR	1	5	5	22	78	100	0
	All	1	1	1	25	75	100	0
D	THR	1	5	5	24	76	100	0
	LT	1	6	6	23	77	100	0
	All	1	6	6	23	77	100	0
Total delay (hours)		0.6						

The HCM results for the roundabout are displayed in Table 5.3. The result shows that 50% of the approaches has a Level of Service (LOS) of A and the other half is grades at B. Moreover, the combined result of LOS archives the grade of B, meaning that the flow is very good. HCM for roundabout details can be found in appendix B

Table 5.3: HCM 2010 results for roundabout layout

Approach	Control delay(veh/hr)	LOS
d.A	13.81	B
d.F	9.83	A
d.D	6.48	A
d.Total	10.04	B

5.3 Comparison of alternatives

Compering the alternative design with the original it is possible to confirm that both suggested alternatives designs improve traffic efficiency. However, the roundabout solutions provide a slightly better traffic flow with a more balanced and lower delay per second for each vehicle. While only being a 0.1-hour difference, the difference reflects meaningful improvements in traffic interaction quality and geometry consistency.

In alternative one, the yield-controlled layout, delay values differ more noticeably between movements. The left turn on approach A and the right turn on approach B experience somewhat higher delays due to conflict gaps and higher conflict friction at the yield line. These movements relay heavily on gap acceptance and small fluctuations in opposing flow can increase hesitation and lead to uneven delay distribution. As a result, the accumulated delay is 0.7 hours even though congestion remains low.

The roundabout alternative shows a uniformed delay pattern, with nearly all movements experienced 5-6 seconds delay per vehicle. Since, the roundabout remove direct conflict movements and replaces them with a continuous flow geometry, vehicles encounter more predictable interactions and shorter yielding times at entry. The circulation geometry flow also regulates entry gaps more consistently and reduces movement specific variability. This results in an accumulated delay of 0.6 which is better than the first alternative with only yielding approaches.

The HCM 2010 operation evaluation for the roundabout further supports the findings. Following the HCM methodology, the average control delay across all approaches is 10.4 seconds per vehicle, corresponding to LOS B, while the individual approaches have a LOS ranging from A-B. These delays reflect short yielding times and stable operating conditions even during peak demands. The HCM result show that the proposed roundabout solution operates within the very good operational range. Comparing the HCM and CAPCAL results each method confirms that the roundabout provides an overall better solution.

The goal of this study was to process collected data from Bobergsgatan and Norra Fiskartorpsvägen intersection. After assessing the performance under the current conditions and proposing and evaluating alternative design solutions to improve the flow and reduce delay, the recommendation is to replace the intersection with a roundabout displayed in Figure 5.2.

Appendix A

$$V_1 = \frac{2 \times 39}{\frac{2 \times 39}{4 \times 22}} = 88$$

$$V_2 = \frac{2 \times 121}{\frac{2 \times 121}{4 \times 61}} = 244$$

Step 1 & 2

	% of Heavy Vehicles	# of Vehicles
(5)	14%	A-B 104 →
(6)	1%	A-C 592 ↘

$$\begin{aligned} \leftarrow 88 \quad F-E \quad 21\% \quad (1) \\ \searrow 244 \quad F-C \quad 0 \quad (2) \end{aligned}$$



⇒ Bobergsgatan (1,2) is the major road

344 128
D-E D-B

8% 18%
(3) (4)

Step 3 conflicting flow rate

Left-Turn Movements for Major-street

$$V_{c,2} = V_5 + V_6 = 104 + 592 = 696 \text{ veh/hr}$$

Right-Turn Movements for Minor-street

$$V_{c,4} = V_5 + 0.5 V_6 = 104 + 296 = 400 \text{ veh/hr}$$

Left-Turn Movements for Minor-street

$$V_{c,3} = V_5 + 0.5 V_6 + 2 V_2 + V_1 = 400 + 2 \times 244 + 88 = 976 \text{ veh/hr}$$

Step 4 Critical headway & followup headway

$$t_{c,x} = t_{c,HV} P_{HV} + t_{c,AG} - t_{3,5s}$$

$$t_{f,x} = t_{f,base} + t_{f,HV} P_{HV}$$

Left-turn for major street

$$t_{c,2} = 4.1 + 1 \times 0 + 0.2 \times 0 - 0 = 4.1 \text{ s}$$

$$t_{f,2} = 2.2 + 0.9 \times 0 = 2.2 \text{ s}$$

Right-turn for minor street

$$t_{c,4} = 6.2 + 1 \times 0.18 + 0.1 \times 0 - 0 = 6.37 \text{ s}$$

$$t_{f,4} = 3.3 + 0.9 \times 0.18 = 3.46 \text{ s}$$

Left-turn for minor street

$$t_{c,3} = 7.1 + 1 \times 0.08 + 0.2 \times 0 - 0.7 = 6.48 \text{ s}$$

$$t_{f,3} = 3.5 + 0.9 \times 0.08 = 3.57 \text{ s}$$

Step 5 Potential capacity

$$C_{p,2} = 696 \times \frac{e^{\frac{696 \times 9.1}{3600}}}{1 - e^{\frac{696 \times 9.1}{3600}}} \approx 895 \text{ veh/hr}$$

$$C_{p,4} = 400 \times \frac{e^{\frac{400 \times 6.37}{3600}}}{1 - e^{\frac{400 \times 6.37}{3600}}} \approx 617 \text{ veh/hr}$$

$$C_{p,3} = 976 \times \frac{e^{\frac{976 \times 6.48}{3600}}}{1 - e^{\frac{976 \times 6.48}{3600}}} = 263 \text{ veh/hr}$$

Step 6 Rank 1 major-street right turn no delay

Step 7 Rank 2

Left-turn for major street $C_{m,2} = C_{p,2} = 895 \text{ veh/hr}$

Left turn for minor street: $C_{m,3} = C_{p,3} \times (1 - \frac{V_2}{C_{m,2}})$

Right-turn for minor street $C_{m,4} = C_{p,4} = 617 \text{ veh/hr}$

$$= 263 \times (1 - \frac{60}{299})$$

$$= 198$$

Step 8 Rank 3

$$C_{m,k} = (C_{p,k}) f_k \quad f_k = \prod_j P_{0,j} \Rightarrow P_{0,j} = 1 - \frac{V_j}{C_{m,j}}$$

$$P_{0,2} = 1 - \frac{244}{895} = 0.73$$

Step 9 Control delay

Rank 2

$$d = \frac{3600}{C_{m,x}} + 900 \left[\frac{1/x}{C_{m,x}} - 1 + \sqrt{\left(\frac{V_x}{C_{m,x}} - 1 \right)^2 + \frac{\left(\frac{3600}{C_{m,x}} \right) \left(\frac{V_x}{C_{m,x}} \right)}{450T}} \right] + 5$$

0.53

$$d_2 = \frac{3600}{895} + 900 \times 0.27 \times \left[-0.73 + \sqrt{\left(\frac{244}{895} - 1 \right)^2 + \frac{\left(\frac{3600}{895} \right) \left(\frac{244}{895} \right)}{450 \times 0.25}} \right] + 5$$

$$= 10.55 \text{ s/veh}$$

$$d_4 = \frac{3600}{617} + 900 \times 0.25 \times \left[-0.79 + \sqrt{\left(\frac{128}{617} - 1 \right)^2 + \frac{\left(\frac{3600}{617} \right) \left(\frac{128}{617} \right)}{450 \times 0.25}} \right] + 5$$

$$= 12.36 \text{ s/veh}$$

$$d_3 = \frac{3600}{198} + 900 \times 0.25 \times \left[0.74 + \sqrt{\left(\frac{344}{198} - 1 \right)^2 + \frac{\left(\frac{3600}{198} \right) \left(\frac{344}{198} \right)}{450 \times 0.25}} \right] + 5$$

$$= 393.4 \text{ s/veh}$$

Appendix B

Table B.1 Adjust flow rates with heavy vehicles

	vi-start	%HV	f.HV	vi,pce
v1-start	88	21%	0.83	106
v2-start	244	0%	1.00	244
v3-start	344	8%	0.93	370
v4-start	128	18%	0.85	151
v5-start	104	14%	0.88	118
v6-start	592	1%	0.99	598

Table B.2 Circulating and exiting movements

v.circle.A,pce	598	Circulating entry North(A)
v.circle.F,pce	370	Circulating entry South(F)
v.circle.D,pce	118	Circulating entry West(D)
v.exit E,pce	476	Exit North (E)
v.exit B,pce	269	Exit South (B)
v.exit.C,pce	842	Exit West (C)

Table B.3 Entry flow rate

v.entry.A.THR,pce	118	North entry (A) - only thru traffic
v.entry.A.RT,pce	598	North entry (A) - only right turning traffic
v.entry.F,pce	350	South entry(F)- Both thru and right turning traffic
v.entry.D.LT,pce	370	West entry (C)- only thru traffic
v.entry.D.RT,pce	151	West entry (C)- only right turning traffic

Table B.4 Capacity of entry lanes

Entering lanes	Conflicting lanes	Entry lane	Capacity(pc/hr)
2	1	C.entry.A.THR,pce	881
2	1	C.entry.A.RT,pce	881
1	1	C.entry.F,pce	780
2	1	C.entry.D.LT,pce	1005
2	1	C.entry.D.RT,pce	1005

Table B.5 Capacity of entry lanes converted to veh/hr

		Entry lane	Capacity(veh/hr)
f.HV	0.88	c.A.THR	1001
f.HV	0.99	c.A.RT	890
f.HV	0.91	c.F	857
f.HV	0.93	c.D.LT	1080
f.HV	0.85	c.D.RT	1182

Table B.6 Volume-to-capacity ratio

x.A.THR	0.12
x.A.RT	0.67
x.F	0.41
x.D.LT	0.34
x.D.RT	0.13

Table B.7 Average control delay and LOS

	Control delay (s/veh)	LOS
d.A.THR	5.24	A
d.A.RT	15.31	B
d.F	9.83	A
d.D.LT	7.12	A
d.D.RT	4.77	A

Table B.8 Average control delay and LOS for each approach and the whole roundabout

Approach	Control delay(veh/hr)	LOS
d.A	13.81	B
d.F	9.83	A
d.D	6.48	A
d.Total	10.04	B

Appendix C

Table C.1: Table 6 Speeds of the vehicles at K1.

No	Time	Vehicle Type	Vehicle Speed at K1
1	13:45:02995	Car	34,48
2	13:45:26769	Car	43,44
3	13:45:47422	Car	35,79
4	13:45:49354	Car	36,67
5	13:46:05671	Van/MPV	34,44
6	13:46:06213	Lorry/Bus	34,89
7	13:46:19275	Car	47,71
8	13:46:31487	Car	38,63
...	...	...	...
844	17:51:22514	Car	38,26

Table C.2: Table 6 Speeds of the vehicles at K2.

Veh, No	Time	Vehicle Type	Vehicle Speed at K2
1	13:45:39827	Car	33,61
2	13:45:42111	Car	35,63
3	13:45:45737	Car	34,84
4	13:45:48660	Car	35,46
6	13:46:01164	Car	34,59
7	13:46:10288	Car	31,35
8	13:46:23746	Car	32,02
...	...	...	...
1657	17:00:22796	Car	29,02

Table C.3: Flow and Mean Speed for each time interval for K1

Time interval	Time	Travel Direction	Number of vehicles	Average speed at K1
1	13:45 - 13:50	1-0	19	36,8
2	13:50 - 13:55	1-0	14	30,6
3	13:55 - 14:00	1-0	13	35,6
4	14:00 - 14:05	1-0	21	34,7
...	...	...	...	...
50	17:50 - 17:55	1-0	8	24,8

Table C.4: Flow and Mean Speed for each time interval for K2

Time interval	Time	Travel Direction	Number of vehicles	Average speed at K2
1	13:45 - 13:50	0-1	34	34,5
2	13:50 - 13:55	0-1	37	32,1
3	13:55 - 14:00	0-1	31	35,2
4	14:00 - 14:05	0-1	25	29,9
...	...	...	...	...
40	17:00 – 17:05	0-1	4	29.6